Experiment Report 1 Which Networks Carry My Packets

CSC6410 | 1 November 2026 | Windows on home network

I traced five web services twice from my Windows computer and mapped visible public router addresses to AS numbers with RIPEstat. A web hostname sometimes led to a hosting network rather than the institution's own AS. My local traceroute mapping and three remote BGP collectors reported the same destination origin AS in the recorded results, but they did not show the same full path.

1 Plan and predictions

Prediction baseline saved at 12:43:58 IST on 1 November 2026, before the first measurement. These estimates were prepared with AI help after earlier discussion of the assignment, so they are not presented as unaided pre-consultation predictions. I used Cloudflare as the fifth site.

Target	IP hops	ASes	Networks expected
www.cuw.edu	14	4	access ISP, transit, CUW or hosting
www.ox.ac.uk	18	5	access ISP, international transit, UK research, Oxford or hosting
www.wikipedia.org	12	3	access ISP, transit, Wikimedia
www.nasa.gov	16	4	access ISP, transit or CDN, NASA or hosting
www.cloudflare.com	10	2	access ISP, Cloudflare

2 Consultation and critique

I asked AI to explain the assignment and help run the Windows experiment. I checked the RIPE NCC documentation for network-info, as-overview, looking-glass, and bgp-state. network-info gives an announcing ASN and containing prefix for an IP; as-overview gives a holder name; looking-glass lists AS paths from remote collectors.

The advice needed corrections. An earlier AI example suggested Google as the fifth site; the actual experiment used Cloudflare, chosen before measurement. A website's organization name is not a reliable prediction of its web server's origin AS: CUW resolved to Amazon, Oxford to Cloudflare, and NASA to Automattic. A public traceroute interface can return no AS from network-info or bgp-state. I kept those hops unknown rather than inventing a hand-off. Collector paths come from other vantage points, so they are not a direct record of my home route.

3 Execution and iterations

Environment: Windows, home network. Each target used tracert -d -h 30 <target>. Run 1 began at 12:44:26 IST and run 2 at 13:46:33 IST, 62.1 minutes later. The raw commands and output are in Appendix 2. I ran python CSC6410_EXP1_Student.py --run run1 and --run run2. The script reads the text, skips non-global addresses, caches RIPE replies, maps ASNs to holder names, and collapses consecutive duplicate ASNs.

I tested the parser on all five saved run 1 traces. I also checked that 10/8, 172.16/12, 192.168/16, and 100.64/10 addresses were skipped while a public IPv4 example was accepted. The script's fallback is to leave a hop unknown when both RIPE views lack a route.

Real issue	Change	Evidence
Public hop without an ASN	Added one bgp-state fallback for network-info misses; left unresolved hops unknown.	CUW hop 11 and Wikipedia hop 13 still had zero fallback routes. Both are marked unmapped.
Privacy mask crashed with no intermediate public hop	Checked that first_public existed before splitting it into octets.	The first rerun raised AttributeError on NoneType; the next rerun completed all five sites.
Institution names did not predict web origin AS	Compared each measured destination IP and its actual origin AS.	CUW AS16509, Oxford AS13335, NASA AS2635; the last visible AS and collector origin agreed.

CUW had 8 timeout rows in run 1 and 8 in run 2. The destination answered in both. A timeout means a router did not reply to this probe; it does not prove forwarding stopped.

Figures 1-3 are actual browser screenshots of RIPEstat responses. The terminal, script, crash before/after, and lab-check screenshots require capture from a visible terminal before submission; the saved text below provides the recorded evidence.


```
{
  "messages": [],
  "see_also": [],
  "version": "1.1",
  "data_call_name": "network-info",
  "data_call_status": "supported",
  "cached": false,
  "query_id": "20261101073409-e348067d-f02e-4b8b-b512-3b61c928bed7",
  "process_time": 13,
  "server_id": "app196",
  "build_version": "v0.12.2-2026.10.23",
  "pipeline": "1478200",
  "status": "ok",
  "status_code": 200,
  "time": "2026-11-01T07:34:09.962893",
  "data": {
    "asns": [
      "14907"
    ],
    "prefix": "103.102.166.0/24"
  }
}
```


Figure 1. RIPEstat network-info returned prefix 103.102.166.0/24 and AS14907 for Wikipedia's destination hop.


```
{
  "messages": [],
  "see_also": [],
  "version": "1.1",
  "data_call_name": "network-info",
  "data_call_status": "supported",
  "cached": false,
  "query_id": "20261101073344-ae073b3c-a1e5-430b-9db8-f51ade885d8d",
  "process_time": 11,
  "server_id": "app196",
  "build_version": "v0.12.2-2026.10.23",
  "pipeline": "1478200",
  "status": "ok",
  "status_code": 200,
  "time": "2026-11-01T07:33:45.008489",
  "data": {
    "asns": [],
    "prefix": ""
  }
}
```


Figure 2. network-info returned an empty ASN list for Wikipedia hop 13.

```
{
  "messages": [],
  "see_also": [],
  "version": "1.2",
  "data_call_name": "bgp-state",
  "data_call_status": "supported",
  "cached": false,
  "query_id": "20261101073207-0cdaf0af-b2e0-4405-b91c-a57cca91377d",
  "process_time": 119,
  "server_id": "app179",
  "build_version": "v0.12.2-2026.10.23",
  "pipeline": "1478200",
  "status": "ok",
  "status_code": 200,
  "time": "2026-11-01T07:32:07.932987",
  "data": {
    "resource": "27.111.228.186",
    "timestamp": "2026-11-01T03:59:45",
    "bgp_state": [],
    "nr_routes": 0
  }
}
```

Figure 3. The bgp-state fallback returned zero routes for the same hop.

4 Results from my measurements

www.cuw.edu

Item	Run 1	Run 2
Destination IP	54.240.162.50	54.240.162.35
IP hops / responding	17 / 9	17 / 9
Visible AS path	AS16509	AS16509
Destination origin	AS16509	AS16509

AS names: AS16509 = AMAZON-02 - Amazon.com, Inc..

RIPE collector	Prefix	AS path
RRC01 London, GB	54.240.162.0/24	48070 61049 174 16509 16509 16509
RRC00 Amsterdam, NL	54.240.162.0/24	59919 9002 6453 16509 16509 16509
RRC24 Montevideo, UY	54.240.162.0/24	46997 38008 38008 174 16509 16509 16509

Last visible ASN matches destination origin: run 1 yes, run 2 yes. Visible run 1 ASes in at least one collector path: AS16509. Absent: none. The collector paths start elsewhere.

www.ox.ac.uk

Item	Run 1	Run 2
Destination IP	172.66.169.161	104.20.34.13
IP hops / responding	11 / 8	11 / 9
Visible AS path	AS13335	AS55836 -> AS13335
Destination origin	AS13335	AS13335

AS names: AS13335 = CLOUDFLARENET - Cloudflare, Inc.; AS55836 = RELIANCEJIO-IN - Reliance Jio Infocomm Limited.

RIPE collector	Prefix	AS path
RRC01 London, GB	172.66.160.0/20	48070 13335
RRC00 Amsterdam, NL	172.66.160.0/20	59919 13335
RRC24 Montevideo, UY	172.66.160.0/20	46997 38008 38008 13335

Last visible ASN matches destination origin: run 1 yes, run 2 yes. Visible run 1 ASes in at least one collector path: AS13335. Absent: none. The collector paths start elsewhere.

www.wikipedia.org

Item	Run 1	Run 2
Destination IP	103.102.166.224	103.102.166.224
IP hops / responding	15 / 13	15 / 13
Visible AS path	AS64049 -> AS14907	AS64049 -> AS14907
Destination origin	AS14907	AS14907

AS names: AS64049 = RJIPL-SG - Reliance Jio Infocomm Pte Ltd Singapore; AS14907 = WIKIMEDIA - Wikimedia Foundation Inc..

RIPE collector	Prefix	AS path
RRC01 London, GB	103.102.166.0/24	48070 61049 174 6453 14907
RRC00 Amsterdam, NL	103.102.166.0/24	59919 9002 6453 14907
RRC24 Montevideo, UY	103.102.166.0/24	46997 38008 38008 3491 6453 14907

Last visible ASN matches destination origin: run 1 yes, run 2 yes. Visible run 1 ASes in at least one collector path: AS14907. Absent: AS64049. The collector paths start elsewhere.

www.nasa.gov

Item	Run 1	Run 2
Destination IP	192.0.66.108	192.0.66.108
IP hops / responding	13 / 10	13 / 10
Visible AS path	AS9583 -> AS2635	AS9583 -> AS2635
Destination origin	AS2635	AS2635

AS names: AS9583 = SIFY-AS-IN - Sify Limited; AS2635 = AUTOMATTIC - Automattic, Inc.

RIPE collector	Prefix	AS path
RRC01 London, GB	192.0.66.0/24	48070 2635
RRC00 Amsterdam, NL	192.0.66.0/24	59919 2635
RRC24 Montevideo, UY	192.0.66.0/24	46997 38008 38008 2635

Last visible ASN matches destination origin: run 1 yes, run 2 yes. Visible run 1 ASes in at least one collector path: AS2635. Absent: AS9583. The collector paths start elsewhere.

www.cloudflare.com

Item	Run 1	Run 2
Destination IP	104.16.123.96	104.16.123.96
IP hops / responding	11 / 9	11 / 9
Visible AS path	AS55836 -> AS13335	AS55836 -> AS13335
Destination origin	AS13335	AS13335

AS names: AS55836 = RELIANCEJIO-IN - Reliance Jio Infocomm Limited; AS13335 = CLOUDFLARENET - Cloudflare, Inc..

RIPE collector	Prefix	AS path
RRC01 London, GB	104.16.112.0/20	48070 13335
RRC00 Amsterdam, NL	104.16.112.0/20	59919 13335
RRC24 Montevideo, UY	104.16.112.0/20	46997 38008 38008 13335

Last visible ASN matches destination origin: run 1 yes, run 2 yes. Visible run 1 ASes in at least one collector path: AS13335. Absent: AS55836. The collector paths start elsewhere.

5 Analysis

Target	Predicted IP / AS	Run 1 IP / AS	Run 2 IP / AS
www.cuw.edu	14 / 4	17 / 1	17 / 1
www.ox.ac.uk	18 / 5	11 / 1	11 / 2
www.wikipedia.org	12 / 3	15 / 2	15 / 2
www.nasa.gov	16 / 4	13 / 2	13 / 2
www.cloudflare.com	10 / 2	11 / 2	11 / 2

In run 1, three IP hop counts exceeded my estimates (CUW, Wikipedia, Cloudflare) and two were below them (Oxford, NASA). The mean was 13.4 observed versus 14 predicted. Visible AS counts were below every AS estimate because many first hops used private addresses, some routers did not reply, and two public interfaces had no RIPE mapping. The AS sequence is a partial observation, not proof of every network crossed.

Oxford's site had only 11 IP hops in run 1 even though the university is on another continent. The destination was in Cloudflare AS13335, so this trace does not show that the packets traveled to Oxford itself. Hop count is not a reliable geographic distance measure with CDN or anycast hosting.

Oxford's DNS answer changed from 172.66.169.161 to 104.20.34.13, both in Cloudflare AS13335. Run 2 also showed Reliance Jio AS55836 before Cloudflare, while run 1 showed only Cloudflare after two timeout rows. Both traces had 11 IP hop positions, but the different DNS answer and visible reply mean this is not proof of a changed BGP route. CUW also resolved to a different Amazon IP in run 2.

In run 1, Cloudflare showed the clearest adjacent AS hand-off: hop 10 mapped to Reliance Jio AS55836 and hop 11 to Cloudflare AS13335. NASA moved from Sify AS9583 at hop 12 to Automattic AS2635 at hop 13. Wikipedia showed Reliance Jio AS64049 at hops 10-12 and Wikimedia AS14907 at hops 14-15, but hop 13 was unmapped, so the exact transition is uncertain. CUW and Oxford showed only the destination AS after private or timed-out hops. This fits the lecture distinction between router replies and BGP reachability. Internal OSPF decisions cannot be recovered from these traces alone.

A policy-related sign is AS-path prepending: Amazon AS16509 appears three times in all three sampled CUW collector paths, and AS38008 appears twice in the Montevideo path. Prepending can affect BGP selection, but the observations do not prove operator intent or the shortest physical route.

6 Pitfalls and limits

Traceroute probes can be filtered or rate limited. DNS and CDN placement can change the destination IP between runs. RIPEstat network-info uses RIS data that can lag the instant of my trace, and an announced interface prefix may not identify the router operator perfectly. If a prefix has multiple origins, this short script displays the first returned ASN. The first visible non-destination public hop was masked as a privacy precaution; no home public IP was positively identified. The raw API cache is local and is not a submission file.

7 Lessons and reflection

I learned to separate responding IP routers, ASes mapped from those replies, and BGP paths seen elsewhere. The most surprising result was that university and government web names ended in hosting companies' ASes. AI helped organize the work and draft the mapper, but I checked its output against the raw traces and RIPE docs and corrected a real NoneType error. Next time I would save my own predictions before any AI help and capture screenshots while each issue is visible.

The script is supplied separately as CSC6410_EXP1_Student.py. A GitHub file URL is unavailable because this machine is not authenticated to a suitable GitHub repository.

Appendix 1 Sources consulted

RIPE NCC documentation accessed 1 November 2026.

RIPE NCC Network Info: https://stat.ripe.net/docs/data-api/api-endpoints/network-info

RIPE NCC AS Overview: https://stat.ripe.net/docs/data-api/api-endpoints/as-overview.html

RIPE NCC Looking Glass: https://stat.ripe.net/docs/data-api/api-endpoints/looking-glass

RIPE NCC BGP State: https://stat.ripe.net/docs/data-api/api-endpoints/bgp-state

AI tool: ChatGPT and Codex, documented in AI Report 1.

Appendix 2 Raw traceroute and script output

The first visible non-destination public hop in each trace has its last two octets masked. Each capture records the command, environment, and time.

run1 www.cuw.edu

```
Environment: Windows, home network
Started: 2026-11-01T12:44:26+05:30
Ended: 2026-11-01T12:46:17+05:30
Command: tracert -d -h 30 www.cuw.edu
Exit code: 0


Tracing route to d27hkn76rnzir7.cloudfront.net [54.240.162.50]
over a maximum of 30 hops:

  1     *        4 ms     *     192.168.29.1
  2     7 ms     5 ms     3 ms  10.10.16.1
  3    13 ms     6 ms     7 ms  172.16.5.10
  4     9 ms     5 ms     7 ms  192.168.247.202
  5     6 ms     6 ms     6 ms  192.168.230.213
  6     7 ms     4 ms    10 ms  192.168.230.194
  7     8 ms     4 ms     9 ms  192.168.59.114
  8     *        *        *     Request timed out.
  9     *        *        *     Request timed out.
 10     *        *        *     Request timed out.
 11    22 ms    17 ms    19 ms  49.44.x.x
 12     *        *        *     Request timed out.
 13     *        *        *     Request timed out.
 14     *        *        *     Request timed out.
 15     *        *        *     Request timed out.
 16     *        *        *     Request timed out.
 17    20 ms    18 ms    20 ms  54.240.162.50

Trace complete.
```

This trace shows responding hops and any timeouts; Section 4 gives the AS mapping.

run1 www.ox.ac.uk

```
Environment: Windows, home network
Started: 2026-11-01T12:44:26+05:30
Ended: 2026-11-01T12:45:12+05:30
Command: tracert -d -h 30 www.ox.ac.uk
Exit code: 0


Tracing route to www.ox.ac.uk.cdn.cloudflare.net [172.66.169.161]
over a maximum of 30 hops:

  1     2 ms     7 ms     *     192.168.29.1
  2    10 ms     8 ms     9 ms  10.10.16.1
  3     9 ms     6 ms     6 ms  172.16.5.8
  4     8 ms     5 ms     5 ms  192.168.247.200
  5     4 ms     6 ms     6 ms  192.168.230.213
  6     5 ms     8 ms     5 ms  192.168.230.195
  7    14 ms     7 ms     5 ms  192.168.59.114
  8     *        *        *     Request timed out.
  9     *        *        *     Request timed out.
 10     *        *        *     Request timed out.
 11    25 ms    19 ms    19 ms  172.66.169.161

Trace complete.
```

This trace shows responding hops and any timeouts; Section 4 gives the AS mapping.

run1 www.wikipedia.org

```
Environment: Windows, home network
Started: 2026-11-01T12:44:26+05:30
Ended: 2026-11-01T12:45:07+05:30
Command: tracert -d -h 30 www.wikipedia.org
Exit code: 0


Tracing route to dyna.wikimedia.org [103.102.166.224]
over a maximum of 30 hops:

  1     7 ms     *        4 ms  192.168.29.1
  2     5 ms     3 ms     8 ms  10.10.16.1
  3     8 ms     4 ms     5 ms  172.16.5.8
  4     9 ms     9 ms     7 ms  192.168.247.200
  5     6 ms     7 ms     6 ms  192.168.230.213
  6     6 ms     6 ms    16 ms  192.168.230.195
  7     8 ms     5 ms    12 ms  192.168.59.116
  8     *        *        *     Request timed out.
  9     *        *        *     Request timed out.
 10    21 ms    16 ms    22 ms  103.198.x.x
 11    96 ms    49 ms    48 ms  103.198.140.89
 12    48 ms    55 ms    52 ms  103.198.140.89
 13    47 ms    48 ms    47 ms  27.111.228.186
 14    54 ms   192 ms    48 ms  103.102.166.157
 15    50 ms    56 ms    53 ms  103.102.166.224

Trace complete.
```

This trace shows responding hops and any timeouts; Section 4 gives the AS mapping.

run1 www.nasa.gov

```
Environment: Windows, home network
Started: 2026-11-01T12:44:26+05:30
Ended: 2026-11-01T12:45:18+05:30
Command: tracert -d -h 30 www.nasa.gov
Exit code: 0


Tracing route to nasa-gov.go-vip.net [192.0.66.108]
over a maximum of 30 hops:

  1     *        4 ms     *     192.168.29.1
  2     7 ms     6 ms     2 ms  10.10.16.1
  3    13 ms     6 ms     9 ms  172.16.5.10
  4    11 ms     5 ms     6 ms  192.168.247.202
  5     9 ms     4 ms     7 ms  192.168.230.213
  6     7 ms     4 ms     9 ms  192.168.230.194
  7     8 ms    10 ms     4 ms  192.168.59.112
  8     *        *        *     Request timed out.
  9     *        *        *     Request timed out.
 10    17 ms    73 ms    21 ms  202.191.x.x
```

```
 11     *        *        *     Request timed out.
 12    17 ms    20 ms    18 ms  202.191.178.34
 13    20 ms    17 ms    18 ms  192.0.66.108

Trace complete.
```

This trace shows responding hops and any timeouts; Section 4 gives the AS mapping.

run1 www.cloudflare.com

```
Environment: Windows, home network
Started: 2026-11-01T12:44:26+05:30
Ended: 2026-11-01T12:45:01+05:30
Command: tracert -d -h 30 www.cloudflare.com
Exit code: 0


Tracing route to www.cloudflare.com [104.16.123.96]
over a maximum of 30 hops:

  1     1 ms     4 ms     7 ms  192.168.29.1
  2     6 ms     3 ms     4 ms  10.10.16.1
  3     6 ms     7 ms     9 ms  172.16.5.8
  4     5 ms     7 ms     7 ms  192.168.247.202
  5     6 ms     5 ms    11 ms  192.168.230.213
  6     6 ms    11 ms     4 ms  192.168.230.194
  7    10 ms     9 ms     9 ms  192.168.59.112
  8     *        *        *     Request timed out.
  9     *        *        *     Request timed out.
 10    17 ms    28 ms     *     49.44.x.x
 11    18 ms    18 ms    12 ms  104.16.123.96

Trace complete.
```

This trace shows responding hops and any timeouts; Section 4 gives the AS mapping.

run2 www.cuw.edu

```
Environment: Windows, home network
Started: 2026-11-01T13:46:33+05:30
Ended: 2026-11-01T13:48:20+05:30
Command: tracert -d -h 30 www.cuw.edu
Exit code: 0


Tracing route to d27hkn76rnzir7.cloudfront.net [54.240.162.35]
over a maximum of 30 hops:

  1     2 ms     3 ms     *     192.168.29.1
  2    13 ms     5 ms     5 ms  10.10.16.1
  3     8 ms     6 ms     5 ms  172.16.5.8
  4     8 ms     7 ms     5 ms  192.168.247.204
  5     8 ms     7 ms     4 ms  192.168.230.213
  6     7 ms     4 ms     6 ms  192.168.230.194
  7     4 ms     6 ms     6 ms  192.168.59.110
  8     *        *        *     Request timed out.
  9     *        *        *     Request timed out.
 10     *        *        *     Request timed out.
 11    15 ms    19 ms    16 ms  49.44.x.x
 12     *        *        *     Request timed out.
 13     *        *        *     Request timed out.
 14     *        *        *     Request timed out.
 15     *        *        *     Request timed out.
 16     *        *        *     Request timed out.
 17    20 ms    16 ms    20 ms  54.240.162.35

Trace complete.
```

This trace shows responding hops and any timeouts; Section 4 gives the AS mapping.

run2 www.ox.ac.uk

```
Environment: Windows, home network
Started: 2026-11-01T13:46:33+05:30
Ended: 2026-11-01T13:47:15+05:30
Command: tracert -d -h 30 www.ox.ac.uk
Exit code: 0


Tracing route to www.ox.ac.uk.cdn.cloudflare.net [104.20.34.13]
over a maximum of 30 hops:
```

```
  1     4 ms     4 ms     *     192.168.29.1
  2    13 ms     5 ms     2 ms  10.10.16.1
  3     8 ms     6 ms     5 ms  172.16.5.10
  4     8 ms     7 ms     6 ms  192.168.247.204
  5     8 ms     7 ms     4 ms  192.168.230.213
  6     7 ms     4 ms     6 ms  192.168.230.195
  7     4 ms     5 ms     5 ms  192.168.59.116
  8     *        *        *     Request timed out.
  9     *        *        *     Request timed out.
 10     *       13 ms     *     49.44.x.x
 11    14 ms    27 ms    24 ms  104.20.34.13

Trace complete.
```

This trace shows responding hops and any timeouts; Section 4 gives the AS mapping.

run2 www.wikipedia.org

```
Environment: Windows, home network
Started: 2026-11-01T13:46:33+05:30
Ended: 2026-11-01T13:47:14+05:30
Command: tracert -d -h 30 www.wikipedia.org
Exit code: 0


Tracing route to dyna.wikimedia.org [103.102.166.224]
over a maximum of 30 hops:

  1     *        7 ms     5 ms  192.168.29.1
  2     8 ms     6 ms     4 ms  10.10.16.1
  3     8 ms     7 ms     6 ms  172.16.5.8
  4     8 ms     7 ms     4 ms  192.168.247.200
  5     7 ms     4 ms     3 ms  192.168.230.213
  6     8 ms     7 ms     6 ms  192.168.230.195
  7    10 ms     4 ms     6 ms  192.168.59.116
  8     *        *        *     Request timed out.
  9     *        *        *     Request timed out.
 10   205 ms   186 ms   188 ms  103.198.x.x
 11   124 ms    47 ms    51 ms  103.198.140.89
 12    49 ms    47 ms    49 ms  103.198.140.89
 13    47 ms    46 ms    46 ms  27.111.228.186
 14    56 ms    48 ms    51 ms  103.102.166.157
 15    50 ms    50 ms    53 ms  103.102.166.224

Trace complete.
```

This trace shows responding hops and any timeouts; Section 4 gives the AS mapping.

run2 www.nasa.gov

```
Environment: Windows, home network
Started: 2026-11-01T13:46:33+05:30
Ended: 2026-11-01T13:47:25+05:30
Command: tracert -d -h 30 www.nasa.gov
Exit code: 0


Tracing route to nasa-gov.go-vip.net [192.0.66.108]
over a maximum of 30 hops:

  1     *        7 ms     *     192.168.29.1
  2     5 ms     3 ms     2 ms  10.10.16.1
  3     6 ms     5 ms     7 ms  172.16.5.10
  4     7 ms     9 ms     5 ms  192.168.247.202
  5     5 ms     7 ms     4 ms  192.168.230.213
  6    14 ms     4 ms     7 ms  192.168.230.194
  7     4 ms     5 ms     6 ms  192.168.59.112
  8     *        *        *     Request timed out.
  9     *        *        *     Request timed out.
 10    16 ms    17 ms    18 ms  202.191.x.x
 11     *        *        *     Request timed out.
 12    24 ms    19 ms    20 ms  202.191.178.34
 13    21 ms    23 ms    16 ms  192.0.66.108

Trace complete.
```

This trace shows responding hops and any timeouts; Section 4 gives the AS mapping.

run2 www.cloudflare.com

```
Environment: Windows, home network
Started: 2026-11-01T13:46:33+05:30
Ended: 2026-11-01T13:47:11+05:30
Command: tracert -d -h 30 www.cloudflare.com
Exit code: 0


Tracing route to www.cloudflare.com [104.16.123.96]
over a maximum of 30 hops:

  1     3 ms     4 ms     *     192.168.29.1
  2    13 ms     5 ms     2 ms  10.10.16.1
  3     8 ms     6 ms     5 ms  172.16.5.8
  4     8 ms     7 ms     5 ms  192.168.247.200
  5     7 ms     3 ms     4 ms  192.168.230.213
  6     5 ms     7 ms     4 ms  192.168.230.195
  7     8 ms     7 ms     5 ms  192.168.59.112
  8     *        *        *     Request timed out.
  9     *        *        *     Request timed out.
 10    24 ms    12 ms     *     49.44.x.x
 11    12 ms    12 ms    13 ms  104.16.123.96

Trace complete.
```

This trace shows responding hops and any timeouts; Section 4 gives the AS mapping.

Script output

```
python CSC6410_EXP1_Student.py --run run1

www.cuw.edu (54.240.162.50): 17 IP hops
Traceroute AS path: AS16509 (AMAZON-02 - Amazon.com, Inc.)
Destination origin ASN(s): AS16509
  RRC01 London, GB: 48070 61049 174 16509 16509 16509
  RRC00 Amsterdam, NL: 59919 9002 6453 16509 16509 16509
  RRC24 Montevideo, UY: 46997 38008 38008 174 16509 16509 16509

www.ox.ac.uk (172.66.169.161): 11 IP hops
Traceroute AS path: AS13335 (CLOUDFLARENET - Cloudflare, Inc.)
Destination origin ASN(s): AS13335
  RRC01 London, GB: 48070 13335
  RRC00 Amsterdam, NL: 59919 13335
  RRC24 Montevideo, UY: 46997 38008 38008 13335

www.wikipedia.org (103.102.166.224): 15 IP hops
Traceroute AS path: AS64049 (RJIPL-SG - Reliance Jio Infocomm Pte Ltd Singapore) -> AS14907 (WIKIMEDIA - Wikimedia Foundation Inc.)
Destination origin ASN(s): AS14907
  RRC01 London, GB: 48070 61049 174 6453 14907
  RRC00 Amsterdam, NL: 59919 9002 6453 14907
  RRC24 Montevideo, UY: 46997 38008 38008 3491 6453 14907

www.nasa.gov (192.0.66.108): 13 IP hops
Traceroute AS path: AS9583 (SIFY-AS-IN - Sify Limited) -> AS2635 (AUTOMATTIC - Automattic, Inc)
Destination origin ASN(s): AS2635
  RRC01 London, GB: 48070 2635
  RRC00 Amsterdam, NL: 59919 2635
  RRC24 Montevideo, UY: 46997 38008 38008 2635

www.cloudflare.com (104.16.123.96): 11 IP hops
Traceroute AS path: AS55836 (RELIANCEJIO-IN - Reliance Jio Infocomm Limited) -> AS13335 (CLOUDFLARENET - Cloudflare, Inc.)
Destination origin ASN(s): AS13335
  RRC01 London, GB: 48070 13335
  RRC00 Amsterdam, NL: 59919 13335
  RRC24 Montevideo, UY: 46997 38008 38008 13335

Saved run1_analysis.json; cache has 27 responses.
python CSC6410_EXP1_Student.py --run run2

www.cuw.edu (54.240.162.35): 17 IP hops
Traceroute AS path: AS16509 (AMAZON-02 - Amazon.com, Inc.)
Destination origin ASN(s): AS16509
  RRC01 London, GB: 48070 61049 174 16509 16509 16509
  RRC00 Amsterdam, NL: 59919 9002 6453 16509 16509 16509
  RRC24 Montevideo, UY: 46997 38008 38008 174 16509 16509 16509

www.ox.ac.uk (104.20.34.13): 11 IP hops
Traceroute AS path: AS55836 (RELIANCEJIO-IN - Reliance Jio Infocomm Limited) -> AS13335 (CLOUDFLARENET - Cloudflare, Inc.)
Destination origin ASN(s): AS13335
  RRC01 London, GB: 48070 13335
  RRC00 Amsterdam, NL: 59919 13335
```

```
  RRC24 Montevideo, UY: 46997 38008 38008 13335

www.wikipedia.org (103.102.166.224): 15 IP hops
Traceroute AS path: AS64049 (RJIPL-SG - Reliance Jio Infocomm Pte Ltd Singapore) -> AS14907 (WIKIMEDIA - Wikimedia Foundation Inc.)
Destination origin ASN(s): AS14907
  RRC01 London, GB: 48070 61049 174 6453 14907
  RRC00 Amsterdam, NL: 59919 9002 6453 14907
  RRC24 Montevideo, UY: 46997 38008 38008 3491 6453 14907

www.nasa.gov (192.0.66.108): 13 IP hops
Traceroute AS path: AS9583 (SIFY-AS-IN - Sify Limited) -> AS2635 (AUTOMATTIC - Automattic, Inc)
Destination origin ASN(s): AS2635
  RRC01 London, GB: 48070 2635
  RRC00 Amsterdam, NL: 59919 2635
  RRC24 Montevideo, UY: 46997 38008 38008 2635

www.cloudflare.com (104.16.123.96): 11 IP hops
Traceroute AS path: AS55836 (RELIANCEJIO-IN - Reliance Jio Infocomm Limited) -> AS13335 (CLOUDFLARENET - Cloudflare, Inc.)
Destination origin ASN(s): AS13335
  RRC01 London, GB: 48070 13335
  RRC00 Amsterdam, NL: 59919 13335
  RRC24 Montevideo, UY: 46997 38008 38008 13335

Saved run2_analysis.json; cache has 31 responses.
```

Iteration command output

```
2026-11-01 IST, Windows home network
File basename normalized to Student for privacy; traceback and result text otherwise follow the captured console output.

Before change:
Command: python .\experiment1\CSC6410_EXP1_Student.py --run run1
Traceback (most recent call last):
  File "D:\New Downloads-2\veeksh\experiment1\CSC6410_EXP1_Student.py", line 181, in <module>
    main()
  File "D:\New Downloads-2\veeksh\experiment1\CSC6410_EXP1_Student.py", line 166, in main
    results = [analyze(target, args.run, cache) for target in TARGETS]
  File "D:\New Downloads-2\veeksh\experiment1\CSC6410_EXP1_Student.py", line 151, in analyze
    pieces = first_public.split(".")
AttributeError: 'NoneType' object has no attribute 'split'

After change:
Command: python .\experiment1\CSC6410_EXP1_Student.py --run run1
www.cuw.edu (54.240.162.50): 17 IP hops
Traceroute AS path: AS16509 (AMAZON-02 - Amazon.com, Inc.)
www.ox.ac.uk (172.66.169.161): 11 IP hops
Traceroute AS path: AS13335 (CLOUDFLARENET - Cloudflare, Inc.)
www.wikipedia.org (103.102.166.224): 15 IP hops
Traceroute AS path: AS64049 (RJIPL-SG - Reliance Jio Infocomm Pte Ltd Singapore) -> AS14907 (WIKIMEDIA - Wikimedia Foundation Inc.)
www.nasa.gov (192.0.66.108): 13 IP hops
Traceroute AS path: AS9583 (SIFY-AS-IN - Sify Limited) -> AS2635 (AUTOMATTIC - Automattic, Inc)
www.cloudflare.com (104.16.123.96): 11 IP hops
Traceroute AS path: AS55836 (RELIANCEJIO-IN - Reliance Jio Infocomm Limited) -> AS13335 (CLOUDFLARENET - Cloudflare, Inc.)
Saved run1_analysis.json; cache has 27 responses.
```

Appendix 3 Lab environment check

```
Windows home network: docker info --format '{{.ServerVersion}}'
failed to connect to the docker API at npipe:////./pipe/dockerDesktopLinuxEngine
```

Docker Desktop's Linux engine was not running, so I checked the available WSL Ubuntu environment.

```
Windows home network: wsl -d Ubuntu -- bash -lc 'uname -r; python3 --version'
6.6.114.1-microsoft-standard-WSL2
Python 3.14.4
```

WSL Ubuntu ran successfully. A cropped screenshot of this visible terminal check still needs to be inserted before submission.